Supplementary Materials for "A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology"

This supplementary materials includes proofs of technical results and additional numerical examples supporting conclusions from Section 4 the main manuscript.

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1 Proofs

1.1 Proof of Theroem 2.2

Proof. Fix a population i and suppress the index i.

Step 1: Reduction to order statistics.

Since F_X is continuous, the latent variables are almost surely distinct, and the order statistics

$$X_{(1)} < \cdots < X_{(N)}$$

are well-defined. By rank-based selection, the observed system consists of indices j such that $r_j \in \mathcal{R}$, yielding observed pairs

$$(Y_{(j)}, X_{(r_j)}), \quad j = 1, \ldots, n.$$

By assumption, for each observed rank j there exists a strictly increasing measurable function ψ_j such that

$$Y_{(j)} = \psi_j(X_{(r_j)}) \quad \text{a.s.}$$

Step 2: Identification via distribution functions.

Fix j. Since ψ_j is strictly increasing, it is invertible on its image. For any $y \in \mathbb{R}$,

$$\mathbb{P}(Y_{(j)} \leq y) = \mathbb{P}(\psi_j(X_{(r_j)}) \leq y) = \mathbb{P}(X_{(r_j)} \leq \psi_j^{-1}(y)).$$

Hence,

$$F_{Y_{(j)}}(y) = F_{X_{(r_j)}}(\psi_j^{-1}(y)),$$

which implies

$$F_{Y_{(j)}}(\psi_j(x)) = F_{X_{(r_j)}}(x) \quad \text{for } F_{X_{(r_j)}}\text{-a.e. } x.$$

Step 3: Quantile representation.

Since F_X is continuous, the distribution of the order statistic $X_{(r_j)}$ is also continuous and strictly increasing on the interior of its support, hence admits a well-defined generalized inverse. Likewise, $F_{Y_{(j)}}$ is continuous and strictly increasing. Thus both admit well-defined inverses.

Let $u = F_{X_{(r_j)}}(x)$. Then

$$F_{Y_{(j)}}(\psi_j(x)) = u,$$

so

$$\psi_j(x) = F^{-1}_{Y_{(j)}}(u) = F^{-1}_{Y_{(j)}}\big(F_{X_{(r_j)}}(x)\big)$$

for $F_{X_{(r_j)}}$-almost every x. Therefore,

$$Y_{(j)} = F^{-1}_{Y_{(j)}} \circ F_{X_{(r_j)}}(X_{(r_j)}) \quad \text{a.s.}$$

Step 4: Uniqueness.

Let $U = F_{X_{(r_j)}}(X_{(r_j)}) \sim \text{Unif}(0,1)$ by the probability integral transform, and define

$$g(u) := \psi_j\big(F^{-1}_{X_{(r_j)}}(u)\big).$$

Then g is strictly increasing and $g(U) \sim F_{Y_{(j)}}$. By the fact that if $U \sim \text{Unif}(0,1)$ and g is increasing, then the distribution of $g(U)$ uniquely determines g almost everywhere, implying $g(u) = F^{-1}_{Y(j)}(u)$ a.e., it follows that

$$g(u) = F^{-1}_{Y_{(j)}}(u) \quad \text{for a.e. } u \in (0,1),$$

and hence

$$\psi_j(x) = F^{-1}_{Y_{(j)}}\big(F_{X_{(r_j)}}(x)\big) \quad \text{for } F_{X_{(r_j)}}\text{-a.e. } x.$$

Thus, the strictly increasing coupling is unique up to null sets.

Step 5: Conclusion.

The representation holds for each observed rank j, and uniqueness implies that the joint distribution of (X, Y) coincides with that induced by the Full House Model. □

1.2 Proof of Lemma 3.2

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}), f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. A similar technique can be applied to the second equation to obtain the result. □

1.3 Proof of Theorem 3.4

Proof. By Lemma 3.2,

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Under the assumption $F_{Y_1} = F_{Y_2} = F_Y$, the distributions of the order statistics admit Beta representations, which allow us to express the mapping $h_{j,k}$ in terms of regularized incomplete beta functions.

$$\begin{aligned}
&\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1} \circ I^{-1}_{r_{1,j}, N_1 - r_{1,j}+1} \circ I_{j, n_1 - j+1} \circ I^{-1}_{k, n_2 - k+1} \circ I_{r_{2,k}, N_2 - r_{2,k}+1} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ \int_{x}^{F_X^{-1} \circ I^{-1}_{r_{1,j}, N_1 - r_{1,j}+1} \circ I_{j, n_1 - j+1} \circ I^{-1}_{k, n_2 - k+1} \circ I_{r_{2,k}, N_2 - r_{2,k}+1} \circ F_X(x)} d\, I_{r_{1,j}, N_1 - r_{1,j}+1}(F_X(y)) \right\} f_{X_{2,(r_{2,k})}}(x)\, dx
\end{aligned}$$

$$
\begin{aligned}
&= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ \; f_{X_{2,(r_{2,k})}}(x)\; dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ \; d\; I_{r_{2,k},N_2-r_{2,k}+1}(F_X(x)) \\
&= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt.
\end{aligned}
$$

This representation holds without additional assumptions. The positive-part operator appears because the integrand may change sign. Under condition (8), the integrand is nonnegative, hence

$$
\begin{aligned}
&\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt \\
&= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\} \; dt.
\end{aligned}
$$

Thus,

$$
\begin{aligned}
&\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt \\
&\quad = \int_0^1 I_{j,n_1-j+1}(x)\, dI_{k,n_2-k+1}(x) - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x)\, dI_{r_{2,k},N_2-r_{2,k}+1}(x) \\
&\quad = \int_0^1 I_{j,n_1-j+1}(x)\, \frac{x^{k-1}(1-x)^{n_2-k}}{B(k,n_2-k+1)}\, dx \\
&\qquad - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x)\, \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k},N_2-r_{2,k}+1)}\, dx \\
&\quad = \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p (1-x)^{n_1-p} \frac{x^{k-1}(1-x)^{n_2-k}}{B(k,n_2-k+1)}\, dx \\
&\qquad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1-p} \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k},N_2-r_{2,k}+1)}\, dx \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p,n_2+n_1-p-k+1)}{B(k,n_2-k+1)} \\
&\qquad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p,N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k},N_2-r_{2,k}+1)}.
\end{aligned}
$$

□

1.4 Proof of Proposition 3.5

Proof. Specification of $1_i(X_{i,j} \geq X_{i,(N_i-n_i+1)})$ implies that $r_{i,j} = N_i + 1 - j$. When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $x \in (0,1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased [Askitis, 2021], or n_2 or N_1 are decreased. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned}
&I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\
&\quad \implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\
&\quad \implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t.
\end{aligned}$$

Technical condition (8) holds and the conclusion of Theorem 3.4 applies. □

1.5 Proof of Proposition 4.2

Proof. First, in combination with the Definition 4.1 and the assumption $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, we have

$$F_{Y_1}(x)^{\frac{n_1}{N_1}} \geq F_{Y_2}(x)^{\frac{n_2}{N_2}} \implies x \geq F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right).$$

By applying the Equation 7 from the Lemma 3.2 we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \mathrm{P}\left(F^{-1}_{Y_1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{-\infty}^{+\infty} \int_{F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x)
\end{aligned}$$

$$= \int_{-\infty}^{+\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_{2,(n_2)}}(x)$$
$$= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt.$$

Based on the Definition 4.1, we have

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\geq \int_0^1 F_{Y_2}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\leq \int_0^1 \left(t^{1/n_2} + \varepsilon\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}.
\end{aligned}$$

□

1.6 Proof of Theorem 4.3

Proof. Equation 7 from Lemma 3.2 yields

$$g_{n_1,n_2}(x) = F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right) = \Phi^{-1}\left(\Phi\left(\frac{x-\mu_2}{\sigma_2}\right)^{\frac{n_2 N_1}{N_2 n_1}}\right)\sigma_1 + \mu_1.$$

Let A and R be as defined, respectively, in (11) and (13). We have,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right)$$

$$
\begin{aligned}
&= \mathrm{P}\left(F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2N_1}{N_2n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_A \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2N_1}{N_2n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x) \\
&= \int_A F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}} dF_{Y_2,(n_2)}(x) - \int_{A^c} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}} dF_{Y_2,(n_2)}(x) + R.
\end{aligned}
$$

We continue with the left-hand side of the last equality,

$$
\begin{aligned}
&\int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) = \int_0^1 \left[F_{Y_1}\left(F_{Y_2}^{-1}(t^{1/n_2})\right)^{n_1} - t^{N_1/N_2}\right] dt \\
&\quad = \int_0^1 \left(\Phi\left(\frac{\Phi^{-1}(t^{1/n_2})\sigma_2 + \mu_2 - \mu_1}{\sigma_1}\right)\right)^{n_1} dt \; - \; \frac{N_2}{N_1+N_2} \\
&\quad = \int_{-\infty}^{+\infty} \Phi\left(\frac{\sigma_2 x + \mu_2 - \mu_1}{\sigma_1}\right)^{n_1} \left(n_2\Phi(x)^{n_2-1}\phi(x)\right) dx \; - \; \frac{N_2}{N_1+N_2} \\
&\quad = n_2 \int_{-\infty}^{+\infty} \mathbb{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1}X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1}X \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 \end{array} \middle| \; X = x \right) \phi(x)\, dx \; - \; \frac{N_2}{N_1+N_2} \\
&\quad = n_2\, \mathbb{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1}X \le \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1}X \le \frac{\mu_2-\mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 \end{array} \right) \; - \; \frac{N_2}{N_1+N_2} \\
&\quad = n_2\, F_{\mathbf{W}}\left(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)\right) \; - \; \frac{N_2}{N_1+N_2}.
\end{aligned}
$$

The distribution function $F_{\mathbf{W}}$ corresponds to random variable $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ where $\mathbf{\Sigma}$ is as defined in (1) and (2). The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2-\mu_1}{\sigma_1}$ and the last $n_2 - 1$ elements are 0.

The bound on the remainder (17) follows from the argument in Section 4.2. □

1.7 Proof of Theorem 4.4

Proof.

$$\mathbf{F}_{\mathbf{W}}(\mathbf{w}) = \mathbb{P}\left(\begin{array}{l} Z_1 + \frac{\sigma_2}{\sigma_1} Z_0 \leq \frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, Z_{n_1} + \frac{\sigma_2}{\sigma_1} Z_0 \leq \frac{\mu_2 - \mu_1}{\sigma_1}, \\ \\ Z_{n_1+1} + Z_0 \leq 0, \ldots, Z_{n_1+n_2-1} + Z_0 \leq 0 \end{array} \right)$$

$$= \mathbb{P}\left(\begin{array}{l} Z_1 \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \ldots, Z_{n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0, \\ \\ Z_{n_1+1} \leq -Z_0, \ldots, Z_{n_1+n_2-1} \leq -Z_0 \end{array} \right)$$

Define $M_{n_1} = \max\{Z_1, Z_2, \ldots, Z_{n_1}\}$, $M_{n_2-1} = \max\{Z_{n_1+1}, Z_{n_1+2}, \ldots, Z_{n_1+n_2-1}\}$, then the CDF of $\mathbf{W}$ essentially computes the joint probability:

$$P(M_{n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0 \text{ and } M_{n_2-1} \leq -Z_0)$$

From extreme value theory, as $n_1 \to \infty$, M_{n_1} sharply concentrates around $\sqrt{2 \ln n_1}$ [Embrechts et al., 2013].

Substitute this into the first inequality:

$$\sqrt{2 \ln n_1} \leq \frac{\mu_2 - \mu_1}{\sigma_1} - \frac{\sigma_2}{\sigma_1} Z_0 \implies Z_0 \leq \frac{\frac{\mu_2 - \mu_1}{\sigma_1} - \sqrt{2 \ln n_1}}{\frac{\sigma_2}{\sigma_1}}$$

As n_1 grows, $\frac{\frac{\mu_2 - \mu_1}{\sigma_1} - \sqrt{2 \ln n_1}}{\frac{\sigma_2}{\sigma_1}}$ becomes a very large negative number. This forces the random variable Z_0 deep into its left tail.

Similarly, as $n_2 \to \infty$, M_{n_2-1} sharply concentrates around $\sqrt{2 \ln(n_2 - 1)}$ [Embrechts et al., 2013].

Substitute this into the second inequality:

$$\sqrt{2 \ln(n_2 - 1)} \leq -Z_0 \implies Z_0 \leq -\sqrt{2 \ln(n_2 - 1)}$$

As n_2 grows, $-\sqrt{2\ln(n_2-1)}$ becomes a very large negative number. This forces the random variable Z_0 deep into its left tail.

Then the CDF of $\mathbf{W}$ essentially computes the probability:

$$\mathbf{P}(Z_0 \leq \frac{\frac{\mu_2-\mu_1}{\sigma_1} - \sqrt{2\ln n_1}}{\frac{\sigma_2}{\sigma_1}}, Z_0 \leq -\sqrt{2\ln(n_2-1)}),$$

as $n_1, n_2 \to \infty$.

This probability is dominated entirely by the tail of Z_0. We can use the Mill's ratio [Mills, 1926] and apply it to standard Gaussian tail approximation, $\Phi(-x) \sim \frac{\phi(x)}{x}$ as $x \to \infty$.

Since

$$\lim_{n_1,n_2\to\infty} \frac{\ln n_2}{\ln n_1} > \left(\frac{\sigma_1}{\sigma_2}\right)^2,$$

the tail behavior of Z_0 is dominated by n_2, then

$$F_{\mathbf{W}}(\mathbf{w}) \approx \Phi(-\sqrt{2\ln(n_2-1)}) \sim \frac{\phi(\sqrt{2\ln(n_2-1)})}{\sqrt{2\ln(n_2-1)}} = \frac{1}{\sqrt{4\pi\ln(n_2-1)}}\left(\frac{1}{n_2-1} + o(\frac{1}{n_2})\right)$$

Let $\mathbf{W}^* \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma^*})$ and covariance matrix

$$\mathbf{\Sigma^*} = \begin{bmatrix} \Sigma^*_{11} & \Sigma^*_{12} \\ \Sigma^*_{21} & \Sigma^*_{22} \end{bmatrix}, \tag{1}$$

defined with the following structure

$$\Sigma_{11}^{*}=\begin{bmatrix}\frac{\sigma_4^2}{\sigma_3^2}+1 & \frac{\sigma_4^2}{\sigma_3^2} & \dots & \frac{\sigma_2^2}{\sigma_3^2}\\ \frac{\sigma_4^2}{\sigma_3^2} & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & \frac{\sigma_4^2}{\sigma_3^2}\\ \frac{\sigma_4^2}{\sigma_3^2} & \dots & \frac{\sigma_4^2}{\sigma_3^2} & \frac{\sigma_4^2}{\sigma_3^2}+1\end{bmatrix}_{n_1\times n_1} \quad \Sigma_{12}^{T*}=\Sigma_{21}^{*}=\begin{bmatrix}\frac{\sigma_4}{\sigma_3} & \dots & \dots & \frac{\sigma_4}{\sigma_3}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_4}{\sigma_3} & \dots & \dots & \frac{\sigma_4}{\sigma_3}\end{bmatrix}_{(n_2-1)\times n_1}$$

$$\Sigma_{22}^{*}=\begin{bmatrix}2 & 1 & \dots & 1\\ 1 & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & 1\\ 1 & \dots & 1 & 2\end{bmatrix}_{(n_2-1)\times(n_2-1)}. \tag{2}$$

and $\mathbf{w}^{*}=(\frac{\mu_4-\mu_3}{\sigma_3},\dots,\frac{\mu_4-\mu_3}{\sigma_3},0,\dots,0)$, where the first n_1 elements of $\mathbf{w}^{*}$ are $\frac{\mu_4-\mu_3}{\sigma_3}$ and the last n_2-1 elements are 0.

Then the $\mathbf{F}_{\mathbf{W}^{*}}(\mathbf{w}^{*})$ is

$$\begin{aligned}\mathbf{F}_{\mathbf{W}^{*}}(\mathbf{w}^{*}) &= \mathbb{P}\left(\begin{array}{l} Z_1+\frac{\sigma_4}{\sigma_3}Z_0\le\frac{\mu_4-\mu_3}{\sigma_3},\ \dots,\ Z_{n_1}+\frac{\sigma_4}{\sigma_3}Z_0\le\frac{\mu_4-\mu_3}{\sigma_3},\\ Z_{n_1+1}+Z_0\le 0,\ \dots,\ Z_{n_1+n_2-1}+Z_0\le 0\end{array}\right)\\ &= \mathbb{P}\left(\begin{array}{l} Z_1\le\frac{\mu_4-\mu_1}{\sigma_3}-\frac{\sigma_4}{\sigma_3}Z_0,\ \dots,\ Z_{n_1}\le\frac{\mu_4-\mu_1}{\sigma_3}-\frac{\sigma_4}{\sigma_3}Z_0,\\ Z_{n_1+1}\le -Z_0,\ \dots,\ Z_{n_1+n_2-1}\le -Z_0\end{array}\right)\end{aligned}$$

Since the μ_3, μ_4, σ_3, and σ_4 satisfy the same assumptions as μ_1, μ_2, σ_1, and σ_2 in the theorem statement, and applying the same technique for $\mathbf{F}_{\mathbf{W}}(\mathbf{w})$, we have

$$F_{\mathbf{W}^{*}}(\mathbf{w}^{*})\approx\Phi(-\sqrt{2\ln(n_2-1)})\sim\frac{\phi(\sqrt{2\ln(n_2-1)})}{\sqrt{2\ln(n_2-1)}}=\frac{1}{\sqrt{4\pi\ln(n_2-1)}}\left(\frac{1}{n_2-1}+o(\frac{1}{n_2})\right)$$

Consider the difference between paradox probabilities using two sets of different parameters, and

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \mid \mu_3, \mu_4, \sigma_3, \sigma_4\right) \\
&\quad - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} \mid \mu_1, \mu_2, \sigma_1, \sigma_2\right) \\
&= n_2(F_{\mathbf{W}^*}(\mathbf{w}^*) - F_{\mathbf{W}}(\mathbf{w})) + (R^* - R) \to 0
\end{aligned}$$

as $n_1, n_2 \to \infty$. where $R^* := R(\mu_3, \mu_4, \sigma_3, \sigma_4, n_1, n_2, N_1, N_2)$ and $R := R(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$ are defined in (13) □

2 Simulations

2.1 Simulation Results for Theorem 4.3

We now show the simulation results of our Normal Distribution example under different configurations of $(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$. We verify Theorem 4.3 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 2000. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (2) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The theoretical results are computed from the Theorem 4.3 using mvtnorm package [Genz and Bretz, 2009] in R.

The results are tabulated in Table 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

n_1	n_2	N_1	N_2	μ_1	μ_2	σ_1	σ_2	Simulation Prob	Theoretical Prob
10	40	400	600	-1	1	1.0	3	0.4000	0.4000099
10	20	400	600	0	0	1.0	3	0.3970	0.3970305
10	40	400	300	-1	1	1.0	3	0.5715	0.5714174
10	20	400	600	0	1	1.5	2	0.3645	0.3647783
10	40	400	600	0	1	1.0	3	0.3995	0.3999245
20	40	400	600	0	1	1.0	3	0.3995	0.3999803
10	40	400	600	-1	0	1.5	3	0.4000	0.3995149
20	40	400	600	0	0	1.5	2	0.2970	0.2962431
20	40	400	600	0	0	1.0	3	0.3750	0.3740938
10	20	400	600	-1	1	1.0	3	0.3990	0.4000059

Table 1: Simulation Results for Normal Example.

References

Dimitris Askitis. Logarithmic concavity of the inverse incomplete beta function with respect to the first parameter. *Dimitris Askitis*, 2021.

Paul Embrechts, Claudia Klüppelberg, and Thomas Mikosch. *Modelling extremal events: for insurance and finance*, volume 33. Springer Science & Business Media, 2013.

Alan Genz and Frank Bretz. *Computation of Multivariate Normal and t Probabilities*. Lecture Notes in Statistics. Springer-Verlag, Heidelberg, 2009. ISBN 978-3-642-01688-2.

John P Mills. Table of the ratio: area to bounding ordinate, for any portion of normal curve. *Biometrika*, 18(3/4):395–400, 1926.